Inductive Definitions

1. Give an inductive definition for the set

   $$X = \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, \ldots\}.$$

   if $n \in X$ then $(nx2) \in X$

2. Give an inductive definition for the set

   $$X = \{1, 3, 9, 27, 81, 243, 729, \ldots\}.$$

   if $n \in X$ then $(nx3) \in X$

3. Give an inductive definition for the set

   $$X = \{0, 1, 4, 13, 40, 121, 364, 1093, 3280, 9841, \ldots\}.$$

   if $n \in$ then $((n3x) + 1) \in X$

4. Give an inductive definition for the set

   $$X = \{2, 3, 4, 6, 9, 13, 19, 28, 42, 63, 94, 141, 211, 316, 474, 711, \ldots\}.$$

   if $n \in$ then $[nx1.5] \in X$ wrapping numbers in [ ] floors the number

5. Give an inductive definition for the set

   $$X = \{3, 5, 13, 45, 173, 685, 2733, 10925, 43693, \sout{1747653, \ldots}\}.$$

   if $n \in X$ then $((nx4) - 7) \in X$ 174765

6. Consider the set $X \subseteq \mathbb{N}$ defined as follows.

   (a) $0 \in X$.
   (b) If $n \in X$ then $(n+4) \in X$ and $(n+5) \in X$.
   (c) Nothing is in X unless its membership can be established from the above.

   Give three elements of $\mathbb{N}$ which are elements of X, and three elements of $\mathbb{N}$ which are not elements of X, explaining for each one why it is or is not an element.

   $4, 5, 8, 9, 10 \in X$
   $1, 2, 3, 6, 7 \notin X$
   1, 2, 3, 6, 7 aren't in X because the inductive definition with the base doesn't allow any numbers 1-3, nor 6-7 because +4 and +5 to the elements never compute to these numbers.

   Give a description of the set X using set building notation.

   $X = \{n \in \mathbb{N} | n = 4a + 5b \text{ for some } a, b \in \mathbb{N}\}$

7. Consider the set $X \subseteq \mathbb{N}$ defined as follows.

   (a) $0 \in X$.
   (b) If $n \in X$ then $(n+3) \in X$ and $(n+7) \in X$. $\{n \in \mathbb{N} | n = 3a + 7b \text{ for some } a, b \in \mathbb{N}\}$
   (c) Nothing is in X unless its membership can be established from the above.

   Give three elements of $\mathbb{N}$ which are elements of X, and three elements of $\mathbb{N}$ which are not elements of X, explaining for each one why it is or is not an element.

   $3, 6, 7, 9, 12, 10, 13 \ldots \in X$

   Give a description of the set X using set building notation.

   $4, 5, 8 \notin X$

   4, 5, 8 aren't in X because adjusting +3 and +7 to the elements (starting from base 0) never compute to these numbers.

8. Consider the set $X \subseteq \mathbb{N}$ defined as follows.

   (a) $1 \in X$.
   (b) If $n \in X$ then $(n+3) \in X$ and $(n \times 2) \in X$.
   (c) Nothing is in X unless its membership can be established from the above.

   Give three elements of $\mathbb{N}$ which are elements of X, and three elements of $\mathbb{N}$ which are not elements of X, explaining for each one why it is or is not an element.

   $1, 2, 4, 5, 7, 8, 10, \ldots \in X$
   $3, 6, 9 \notin X$

   Give a description of the set X using set building notation.

   $X = \{n \in \mathbb{N} | n \text{is not divisible by } 3\}$

   because n + 3, and when you do n * 2, you'll never get a number that's divisible by 3, because the base number (1) + 3 isn't a multiple of 3.

9. Consider the set $X \subseteq \mathbb{N}$ defined as follows.

   (a) $4 \in X$ and $7 \in X$.
   (b) If $n \in X$ and $k \in X$ then $(n+k) \in X$.
   (c) Nothing is in X unless its membership can be established from the above.

   Give three elements of $\mathbb{N}$ which are elements of X, and three elements of $\mathbb{N}$ which are not elements of X, explaining for each one why it is or is not an element.

   $4, 7, 8, 11, 12, 15, 18, 16, 19, \ldots \in X$
   $5, 9, 13 \notin X$

   Give a description of the set X using set building notation.

   $\{n \in \mathbb{N} | n = 4a + 7b \text{ for some } a, b \in \mathbb{N}\}$

   your base number(s) 4, 7 are already $\in X$, so 4+4... 4+7, 7+4...7+4... etc

10. Consider the set $X \subseteq \mathbb{N}$ defined as follows.

    (a) $0 \in X$.
    (b) If $n \in X$ and n is even then $(n+1) \in X$; and
        if $n \in X$ and n is odd then $2n \in X$.
    (c) Nothing is in X unless its membership can be established from the above.

    Give three elements of $\mathbb{N}$ which are elements of X, and three elements of $\mathbb{N}$ which are not elements of X, explaining for each one why it is or is not an element.

    $0, 1, 2, 3, 6, 7, 14, 15, 30, 31, 62, 63 \ldots \in X$
    $4, 5, 20, 35 \ldots \notin X$

    Give a description of the set X using set building notation.

    $X = \{n \in \mathbb{N} | n = 2^2 - 1 \text{ or } n = 2^2 - 2 \text{ for some } k \geq 1\}$

    Every element of X is either an odd one $2^2 - 1$ or an even one $2^2 - 2$

11. Describe the set S defined as the smallest set satisfying the following:

    (a) $0 \in S$.
    (b) if $w \in S$ then $1w11 \in S$.

    $S = \{0, 1011, 1101111, 1110111111, \ldots\}$
    S is the set of all strings made of n 1s, followed by a single 0, followed by $2n$ 1s, for any $n \geq 0$
    $S = \{1^n 0 1^{2n} | n \in \mathbb{N}\}$
    where 1^n means n 1s written in a row.

12. Describe the set P defined as the smallest set satisfying the following:

    (a) $\{\varepsilon, 0, 1\} \subseteq P$.
    (b) if $w \in P$ then $\{0w0, 1w1\} \subseteq P$.

$P = \{\varepsilon, 0, 1, 00, 11, 000, 010, 101, 111, 0000, 0110, 1001, 1111, \ldots\}$

$P = \{w \in \{0,1\}^* | w \text{ reads the same forwards and backwards}\}$